

TABLE III. Errors on the V5Z/MP2 bond lengths, the V5Z/MP2 vibrational frequencies, and V5Z/CCSD(T) binding energies for 26 dimers using our pseudopotentials and those by Trail and Needs (TN). The errors are given with respect to the corresponding all-electron values computed at the same level of theory using the DKH Hamiltonian to account for relativistic effects.

Dimer	Bond length ($\text{\AA} \cdot 10^{-2}$)		Frequency (cm^{-1})		E_{binding} (kcal/mol)	
	TN	This work	TN	This work	TN	This work
LiH	2.84	-0.77	-30.10	-12.00	2.28	0.02
BeH	-0.60	-0.50	1.70	2.85	-0.32	-0.09
CH	0.08	-0.11	-10.6	-5.61	0.55	0.21
NH	0.07	-0.04	-8.24	-5.91	0.62	0.40
OH	0.08	-0.01	-7.54	-4.90	0.52	0.27
HF	0.07	-0.01	-5.96	-4.31	0.48	0.15
HCl	-0.22	-0.51	9.75	5.81	-0.61	-0.50
Li ₂	1.55	-1.31	-6.09	-0.80	0.86	-0.33
LiF	3.51	-3.59	-36.33	-22.71	4.60	-1.28
CN	-0.04	-0.42	-27.04	-20.22	1.98	-0.52
CO	0.12	-0.34	-12.27	-2.88	1.78	-0.94
N ₂	0.09	-0.24	-13.14	-6.77	1.96	-0.24
NO	-0.01	-0.19	34.55	-8.05	2.01	0.28
O ₂	0.33	-0.01	-11.86	0.32	1.49	-0.06
F ₂	0.35	-0.08	-4.52	2.91	0.66	-0.18
Na ₂	-1.59	-1.43	-0.10	-0.51	-0.25	-0.18
Si ₂	-0.94	-1.22	4.77	1.77	-0.77	-0.64
P ₂	-0.97	-1.29	9.14	4.59	-0.83	-0.41
S ₂	-1.04	-1.45	7.38	5.41	-1.17	-0.92
Cl ₂	-0.75	-1.44	2.04	2.48	-0.45	-0.74
NaCl	-2.60	-1.50	4.12	-2.14	-1.33	0.26
SiO	-0.35	-0.97	6.88	-3.07	-2.05	-1.75
CS	-0.32	-0.86	2.36	2.59	-0.22	-1.17
SO	-0.75	-1.07	8.93	6.14	-1.14	-1.56
ClO	-0.47	-1.45	5.03	7.40	-0.12	-1.09
CIF	-0.27	-0.87	-0.31	0.30	0.29	-0.48
MAD	0.7	0.8	10	5	1.1	0.5

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CH	0.08	-0.11	-10.6	-5.61	0.55	0.21
NH	0.07	-0.04	-8.24	-5.91	0.62	0.40
OH	0.08	-0.01	-7.54	-4.90	0.52	0.27
HF	0.07	-0.01	-5.96	-4.31	0.48	0.15
HCl	-0.22	-0.51	9.75	5.81	-0.61	-0.50
Li ₂	1.55	-1.31	-6.09	-0.80	0.86	-0.33
LiF	3.51	-3.59	-36.33	-22.71	4.60	-1.28
CN	-0.04	-0.42	-27.04	-20.22	1.98	-0.52
CO	0.12	-0.34	-12.27	-2.88	1.78	-0.94
N ₂	0.09	-0.24	-13.14	-6.77	1.96	-0.24
NO	-0.01	-0.19	34.55	-8.05	2.01	0.28
O ₂	0.33	-0.01	-11.86	0.32	1.49	-0.06
F ₂	0.35	-0.08	-4.52	2.91	0.66	-0.18
Na ₂	-1.59	-1.43	-0.10	-0.51	-0.25	-0.18
Si ₂	-0.94	-1.22	4.77	1.77	-0.77	-0.64
P ₂	-0.97	-1.29	9.14	4.59	-0.83	-0.41
S ₂	-1.04	-1.45	7.38	5.41	-1.17	-0.92
Cl ₂	-0.75	-1.44	2.04	2.48	-0.45	-0.74
NaCl	-2.60	-1.50	4.12	-2.14	-1.33	0.26
SiO	-0.35	-0.97	6.88	-3.07	-2.05	-1.75
CS	-0.32	-0.86	2.36	2.59	-0.22	-1.17
SO	-0.75	-1.07	8.93	6.14	-1.14	-1.56
ClO	-0.47	-1.45	5.03	7.40	-0.12	-1.09
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NH	0.07	-0.04	-8.24	-5.91	0.62	0.40
OH	0.08	-0.01	-7.54	-4.90	0.52	0.27
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CN	-0.04	-0.42	-27.04	-20.22	1.98	-0.52
CO	0.12	-0.34	-12.27	-2.88	1.78	-0.94
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Cl ₂	-0.75	-1.44	2.04	2.48	-0.45	-0.74
NaCl	-2.60	-1.50	4.12	-2.14	-1.33	0.26
SiO	-0.35	-0.97	6.88	-3.07	-2.05	-1.75
CS	-0.32	-0.86	2.36	2.59	-0.22	-1.17
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CN	-0.04	-0.42	-27.04	-20.22	1.98	-0.52
CO	0.12	-0.34	-12.27	-2.88	1.78	-0.94
N ₂	0.09	-0.24	-13.14	-6.77	1.96	-0.24
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Cl ₂	-0.75	-1.44	2.04	2.48	-0.45	-0.74
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CS	-0.32	-0.86	2.36	2.59	-0.22	-1.17
SO	-0.75	-1.07	8.93	6.14	-1.14	-1.56
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Na ₂	-1.59	-1.43	-0.10	-0.51	-0.25	-0.18
Si ₂	-0.94	-1.22	4.77	1.77	-0.77	-0.64
P ₂	-0.97	-1.29	9.14	4.59	-0.83	-0.41
S ₂	-1.04	-1.45	7.38	5.41	-1.17	-0.92
Cl ₂	-0.75	-1.44	2.04	2.48	-0.45	-0.74
NaCl	-2.60	-1.50	4.12	-2.14	-1.33	0.26
SiO	-0.35	-0.97	6.88	-3.07	-2.05	-1.75
CS	-0.32	-0.86	2.36	2.59	-0.22	-1.17
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S ₂	-1.04	-1.45	7.38	5.41	-1.17	-0.92
Cl ₂	-0.75	-1.44	2.04	2.48	-0.45	-0.74
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CS	-0.32	-0.86	2.36	2.59	-0.22	-1.17
SO	-0.75	-1.07	8.93	6.14	-1.14	-1.56
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Dimer	Bond length ($\text{\AA} \cdot 10^{-2}$)		Frequency (cm^{-1})		E_{binding} (kcal/mol)	
	TN	This work	TN	This work	TN	This work
LiH	2.84	-0.77	-30.10	-12.00	2.28	0.02
BeH	-0.60	-0.50	1.70	2.85	-0.32	-0.09
CH	0.08	-0.11	-10.6	-5.61	0.55	0.21
NH	0.07	-0.04	-8.24	-5.91	0.62	0.40
OH	0.08	-0.01	-7.54	-4.90	0.52	0.27
HF	0.07	-0.01	-5.96	-4.31	0.48	0.15
HCl	-0.22	-0.51	9.75	5.81	-0.61	-0.50
Li ₂	1.55	-1.31	-6.09	-0.80	0.86	-0.33
LiF	3.51	-3.59	-36.33	-22.71	4.60	-1.28
CN	-0.04	-0.42	-27.04	-20.22	1.98	-0.52
CO	0.12	-0.34	-12.27	-2.88	1.78	-0.94
N ₂	0.09	-0.24	-13.14	-6.77	1.96	-0.24
NO	-0.01	-0.19	34.55	-8.05	2.01	0.28
O ₂	0.33	-0.01	-11.86	0.32	1.49	-0.06
F ₂	0.35	-0.08	-4.52	2.91	0.66	-0.18
Na ₂	-1.59	-1.43	-0.10	-0.51	-0.25	-0.18
Si ₂	-0.94	-1.22	4.77	1.77	-0.77	-0.64
P ₂	-0.97	-1.29	9.14	4.59	-0.83	-0.41
S ₂	-1.04	-1.45	7.38	5.41	-1.17	-0.92
Cl ₂	-0.75	-1.44	2.04	2.48	-0.45	-0.74
NaCl	-2.60	-1.50	4.12	-2.14	-1.33	0.26
SiO	-0.35	-0.97	6.88	-3.07	-2.05	-1.75
CS	-0.32	-0.86	2.36	2.59	-0.22	-1.17
SO	-0.75	-1.07	8.93	6.14	-1.14	-1.56
ClO	-0.47	-1.45	5.03	7.40	-0.12	-1.09
ClF	-0.27	-0.87	-0.31	0.30	0.29	-0.48
MAD	0.7	0.8	10	5	1.1	0.5

TABLE III. Errors on the V5Z/MP2 bond lengths, the V5Z/MP2 vibrational frequencies, and V5Z/CCSD(T) binding energies for 26 dimers using our pseudopotentials and those by Trail and Needs (TN). The errors are given with respect to the corresponding all-electron values computed at the same level of theory using the DKH Hamiltonian to account for relativistic effects.

Dimer	Bond length ($\text{\AA} \cdot 10^{-2}$)		Frequency (cm^{-1})		E_{binding} (kcal/mol)	
	TN	This work	TN	This work	TN	This work
LiH	2.84	-0.77	-30.10	-12.00	2.28	0.02
BeH	-0.60	-0.50	1.70	2.85	-0.32	-0.09
CH	0.08	-0.11	-10.6	-5.61	0.55	0.21
NH	0.07	-0.04	-8.24	-5.91	0.62	0.40
OH	0.08	-0.01	-7.54	-4.90	0.52	0.27
HF	0.07	-0.01	-5.96	-4.31	0.48	0.15
HCl	-0.22	-0.51	9.75	5.81	-0.61	-0.50
Li ₂	1.55	-1.31	-6.09	-0.80	0.86	-0.33
LiF	3.51	-3.59	-36.33	-22.71	4.60	-1.28
CN	-0.04	-0.42	-27.04	-20.22	1.98	-0.52
CO	0.12	-0.34	-12.27	-2.88	1.78	-0.94
N ₂	0.09	-0.24	-13.14	-6.77	1.96	-0.24
NO	-0.01	-0.19	34.55	-8.05	2.01	0.28
O ₂	0.33	-0.01	-11.86	0.32	1.49	-0.06
F ₂	0.35	-0.08	-4.52	2.91	0.66	-0.18
Na ₂	-1.59	-1.43	-0.10	-0.51	-0.25	-0.18
Si ₂	-0.94	-1.22	4.77	1.77	-0.77	-0.64
P ₂	-0.97	-1.29	9.14	4.59	-0.83	-0.41
S ₂	-1.04	-1.45	7.38	5.41	-1.17	-0.92
Cl ₂	-0.75	-1.44	2.04	2.48	-0.45	-0.74
NaCl	-2.60	-1.50	4.12	-2.14	-1.33	0.26
SiO	-0.35	-0.97	6.88	-3.07	-2.05	-1.75
CS	-0.32	-0.86	2.36	2.59	-0.22	-1.17
SO	-0.75	-1.07	8.93	6.14	-1.14	-1.56
ClO	-0.47	-1.45	5.03	7.40	-0.12	-1.09
ClF	-0.27	-0.87	-0.31	0.30	0.29	-0.48
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CH	0.08	-0.11	-10.6	-5.61	0.55	0.21
NH	0.07	-0.04	-8.24	-5.91	0.62	0.40
OH	0.08	-0.01	-7.54	-4.90	0.52	0.27
HF	0.07	-0.01	-5.96	-4.31	0.48	0.15
HCl	-0.22	-0.51	9.75	5.81	-0.61	-0.50
Li ₂	1.55	-1.31	-6.09	-0.80	0.86	-0.33
LiF	3.51	-3.59	-36.33	-22.71	4.60	-1.28
CN	-0.04	-0.42	-27.04	-20.22	1.98	-0.52
CO	0.12	-0.34	-12.27	-2.88	1.78	-0.94
N ₂	0.09	-0.24	-13.14	-6.77	1.96	-0.24
NO	-0.01	-0.19	34.55	-8.05	2.01	0.28
O ₂	0.33	-0.01	-11.86	0.32	1.49	-0.06
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S ₂	-1.04	-1.45	7.38	5.41	-1.17	-0.92
Cl ₂	-0.75	-1.44	2.04	2.48	-0.45	-0.74
NaCl	-2.60	-1.50	4.12	-2.14	-1.33	0.26
SiO	-0.35	-0.97	6.88	-3.07	-2.05	-1.75
CS	-0.32	-0.86	2.36	2.59	-0.22	-1.17
SO	-0.75	-1.07	8.93	6.14	-1.14	-1.56
ClO	-0.47	-1.45	5.03	7.40	-0.12	-1.09
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